

Mitigating renewables curtailment and carbon emissions in California through water sector demand flexibility

Amanda M. Rupiper^a, Robert Good^b, Gregory J. Miller^c, Frank Loge^{b,*}

^a Department of Chemical and Environmental Engineering, University of California, Riverside, CA, USA

^b Department of Civil and Environmental Engineering, University of California, Davis, CA, 95616, USA

^c Energy Systems, Energy and Efficiency Institute, University of California, Davis, CA, 95616, USA

ARTICLE INFO

Handling Editor: Cecília Maria Villas Bôas de Almeida

Keywords:

Energy load shifting
Electric rate tariffs
Demand response
Renewables integration
Water distribution

ABSTRACT

Increased renewable energy deployment in places like California have led to more frequent curtailment of these resources when their generation exceeds simultaneous demand for electricity, resulting in wasted opportunities for electric grid decarbonization. Public potable water distribution systems, which account for approximately 1.2% of statewide electricity demand, may be able to shift the timing of their operations by filling their storage tanks when renewable energy is prevalent to help the California energy grid reduce renewables curtailment, grid emissions, and peak energy demand. We simulate the water demand, pumping, and energy demands of 702 public water distribution systems serving 84% of Californian residents under eight different energy pricing and emissions minimization scenarios. We find that the potable water distribution sector in California has the flexibility to shift its energy demands by up to 1071 GWh annually, reduce renewable energy curtailment by up to 68%, reduce the net demand peak by up to 321 MW, and avoid 330,627 mTCE of emissions. The scenarios that led to the greatest load shifting and emissions reductions included time-of-use energy rates that incentivize consumption during the middle of the day without a demand charge. The findings of this work not only highlight the potential of the water sector to help meet future energy and emissions goals, but also how the energy sector can best incentivize this shift.

1. Introduction

To prevent the worst of global climate change impacts, the United Nations have set goals to reduce carbon emissions by 45% by 2030 and to zero net emissions by 2050. In spite of these ambitious goals, countries around the world are falling short of their emissions targets (United Nations, 2021). The United States is currently the second largest greenhouse gas emitting country in the world, but states like California have taken on aggressive targets to reduce their emissions and increase their consumption of renewable resources (Ge et al., 2020). California aims to reach 100 percent renewable zero carbon energy by 2045.

The ongoing transition from historically fossil fuel powered electric grids toward increased integration of renewable and low carbon energy sources has created new challenges for electric grid operation (Denholm and Hand, 2011). Some renewable energy sources are variable and intermittent, resulting in potential misalignment between the timing of renewable energy generation and electricity demand. The addition of more renewable energy sources in California each year have increased

the frequency of occasions where anticipated generation exceeds the demand for electricity at that same time resulting in oversupply. To prevent over generation, excess energy is frequently curtailed by reducing output to match demand (USEIA, 2021). In 2020, California curtailed more than 1500 Gigawatt-hours (GWh) of solar and wind generation due to projected supply exceeding demand (CAISO, 2021a). Increased deployment of renewable energy in the future is likely to exacerbate this problem resulting in wasted opportunities for grid decarbonization. Without massive investments in energy storage or adjusting energy demands to match the renewable supply, California will not be able to reach their 100 percent renewable energy goal.

As more countries world-wide strive to make aggressive carbon reductions by increasing their reliance on renewable energy, mismatched generation and demand timing will likely limit the ability to integrate sufficient renewable resources to meet emissions goals. In order to minimize renewables curtailment, multiple solutions have been proposed including storing energy that would otherwise have been curtailed and demand response, which shifts demand to utilize renewable

* Corresponding author.

E-mail address: fjloge@ucdavis.edu (F. Loge).

<https://doi.org/10.1016/j.jclepro.2023.140373>

Received 14 July 2022; Received in revised form 18 December 2023; Accepted 22 December 2023

Available online 27 December 2023

0959-6526/© 2023 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

resources when they are available (Zohrabian et al., 2021). One form of demand response, load shifting, moves energy load to alternate time periods and has emerged as a method to increase grid reliability and assimilation of renewable resources (Park et al., 2019; Wimpler et al., 2017). Sectors that have existing flexible production schedules, batch processes or storage, such as plug-in electric vehicles and water pumping and storage, have great potential to perform demand response in the short term (Aghajanzadeh et al., 2015; Kernan et al., 2017; Park et al., 2019; Van Triel and Lipman, 2020; Amaral Lopes et al., 2020; Zohrabian et al., 2021).

In spite of identifying the promise of the water sector as a load shift resource, research in this space has been limited to single entity case studies and there have not yet been any attempts to quantify the broader scale of this opportunity or the challenges, such as in California as a whole using historical energy rates (Kernan et al., 2017; Oikonomou et al., 2018; Park et al., 2019; Amaral Lopes et al., 2020; Sowby, 2023). Mobilizing the water sector to shift when it consumes energy may require changing how energy utilities incentivize energy consumption (Zohrabian et al., 2021). Historically, energy pricing structures have been built for a fossil fuel based market. As alternative energy sources enter the mix, old incentives may no longer be achieving their intended goals.

Utilizing a mass-balance model of water distribution system operations, paired with a pump scheduling algorithm, this study seeks to estimate the statewide potential for load shifting resources from California's water distribution systems under different goals and pricing scenarios. The objective of this research is to estimate how much energy demand flexibility is realistically available from California's water distribution, the impacts of this flexibility on grid emissions and renewable integration, and identify how the electricity sector could most effectively incentivize beneficial load shifting through their rate structures.

While using California as a large case study of many water utilities, this research demonstrates the potential demand flexibility of any water system that has water storage facilities and how the water sector in other states or countries could play a sizable role toward greening the energy sector and achieving global sustainable development goals. The findings of this research are useful to water suppliers aiming to reduce their energy costs and carbon footprints as well as to energy suppliers and energy rate designers seeking to improve grid reliability and integrate greater quantities of renewable resources on the grid.

2. Background

Water distribution systems are responsible for reliably delivering safe water to end users at an adequate pressure, accomplished through a combination of pumps, storage reservoirs, and gravity. Like an electrical system, supply and demand must always be balanced to maintain system pressure. Closed pressure zones, which do not contain any water storage, may require variable speed pumps which ramp their flows up and down to match the real-time water demands in that zone. Pumping too little water will drop the pressure, and pumping too much can over pressurize the system, leading to damage. For this reason, many parts of a water system are in open pressure zones, which contain storage facilities. In these open zones, pumping water at a rate greater than demand fills the water reservoir. Turning pumps off or operating below demand causes the water storage tanks to drain and provide pressure to the system. From the standpoint of the flexibility of pump energy timing, closed pressure zones have no flexibility; the only way to modify the load shape of these pumps would be to modify the timing of when end users consume water. In open pressure zones, the primary factor that determines flexibility is the amount of water storage available relative to water demand and pumping capacity. With enough storage volume, pumps could be run to fill the reservoirs at one time of day and turned off at another time of day to meet demand using the water retained in the storage reservoirs.

How well the flexibility within the water sector aligns with the needs

of the electric grid depends on the electric rate structure. Electricity can be the single largest operating cost of a water distribution system and pumps will be operated to minimize these costs to the extent that they are still able to reliably supply safe water. Electric rates are often made up of two primary factors: a consumption charge (per kWh), and a demand charge for peak consumption interval (per kW). Consumption charges account for the actual energy used and demand charges account for the generation capacity required to meet the peak demands of the system (Sherwood et al., 2016). High demand charges may encourage utilities to pump at constant rates to keep any one moment of demand at a minimum. Time-of-use (TOU) consumption charges, with variable pricing dependent on the time of the day, may encourage water utilities to shift when they consume energy to take advantage of low energy prices or avoid high costs at a different time (Faruqui and Sergici, 2013). Many electric utilities offer several different rate structures for a consumer to select from, which may be comprised of different types of demand and consumption rates.

Due to the flexibility of water distribution systems and depending on the energy rate structures available to them, the water sector may be able to permanently change its electric load shape as well as perform dynamic load shifting (Menke, et al., 2016; Kernan et al., 2017). Modern fixed TOU energy rates provided by electric utilities typically encourage energy consumption either when renewables are abundant in the middle of the day and/or discourage consumption during peak energy demands in the early evening. The energy rate motivates users to alter when they use energy resulting in a permanent change to their energy load profile, known as load shaping (Faruqui and Sergici, 2013). Water utilities may also have the potential to participate in dynamic load shifting and respond to the daily needs of the electric grid if given access to real-time-pricing (RTP) (Kernan et al., 2017). RTP represents the actual cost of energy supply operations at any given time of day due to the combination of energy resources (fossil and renewable) on the grid at that moment. Cheaper prices typically coincide with a high prevalence of renewable resources on the grid and more expensive prices during peak energy demand periods because of expensive peaking plants being brought online to meet peak demand (Beck and Martinot, 2004). If there is an oversupply of energy available, grid operators will curtail renewable resources and sell energy at very low or even negative prices to balance the grid (CAISO, 2017). Water system operation is often forecasted and planned hours to days in advance, so unlocking the full potential of dynamic load shifting would likely require sophisticated decision support software that could help operators co-optimize these dynamic prices with the other operational needs of the system.

The water distribution sector is listed as one of the most promising short-term opportunities for mitigating oversupply of renewable energy for several reasons (Park et al., 2019). Water storage and pumping infrastructure is already built and integrated with SCADA control systems that could utilize software to make operational changes driven by price signals. Additionally, operators are relatively indifferent to when pumps operate as long as they are able to maintain system reliability, which is not the case in other sectors such as residential, where behavioral barriers can limit flexibility. However, risk aversion, hourly and seasonal variability in water demand, complex energy rate tariffs, and poor belief in the economic advantages are all challenges that have been identified as preventing the water sector from more fully participating in demand response programs (Sowby, 2021).

3. Methods

3.1. Approach

This study models 702 individual public agencies serving potable water to 84% of Californian residents, responsible for water quality, storage, and pressurization of water distribution systems. Simplified hydraulic models were created for each individual utility based on publicly reported data and assumptions from literature. The model was

optimized for each agency to determine the ideal water pumping strategies for several scenarios and aggregated to the statewide scale to demonstrate the flexibility of the water sector as a whole to shift demand in response to pricing signals and aid in achieving statewide load-shifting and GHG emission reduction goals.

3.2. Simplified hydraulic model

Typical hydraulic models are used to represent detailed water distribution systems including pipe networks, pressure zones, storage facilities, pumping stations, and demand nodes. These models can predict hydraulic operation of a water system including pressures, energy consumption, pumping rates, and hydraulic head throughout the modeled region (Robinson et al., 2012). In order to represent the potable water distribution systems of all utilities throughout California it was necessary to make assumptions and simplifications to traditional hydraulic models. These assumptions include: 1) water demand and distribution are represented by a single demand center and single pressure zone, 2) water storage is represented by a single aggregated tank volume within the same pressure zone, 3) pipe networks are ignored, and 4) pumping is represented by a single aggregated pump station, which can operate at variable or continuous speeds.

The simplified model for each utility is shown in Fig. 1, which highlights the primary components of the model as well as the static characteristics applied to each component such as total storage volume ($V_{\max}$) and maximum pump rate ($F_{\max}$). These characteristics were estimated for each utility using data such as individual utility water sales and water storage data extracted from multiple years of California electronic annual reports (eAR) (CA SWRCB, 2023). Additional details about the data used for each modeling component and the methods used to generate them can be found in the supplemental information (SI) section.

The model was used to conduct a mass balance on the water within the storage tank such that volume in the tank at any time increment is equal to the volume at the last time increment (V_{h-1}) minus the water used to meet demand (D_h) plus the water pumped in (F_h) as shown in Equation (1). A pump scheduling algorithm was used to optimize the system such that costs were minimized, taking into account variable energy prices throughout the day and different pricing schemes. The algorithm allowed for changes in pump timing and rate to minimize the cost while constraining the solutions to satisfy the mass balance in Equation (1).

$$V_h = V_{h-1} + F_h - D_h$$

Equation 1

Where:

$0 \leq F_h \leq F_{\max}$ (Pump flow must not exceed pump capacity).

$V_{\min} \leq V_h \leq V_{\max}$ (Storage volume must not exceed capacity or fall below minimum requirements).

F_h = pump flow (MG) during hour h .

V_h = Volume of water (MG) in storage tank at the end of hour h .

D_h = Water distribution demand (MG) during hour h .

3.3. Optimization process

We utilized dynamic programming optimization to determine the optimal pump rate and timing to minimize costs while obeying constraints. This approach solves for the optimum series of controls by starting at the final time-step and evaluating all possible decisions in reverse until the current time-step is met. An initial penalty is provided for any scenario that would result in final storage levels less than the initial storage level (assumed as the average of $V_{\max}$ and $V_{\min}$). The total performance of potential decisions was evaluated as the accumulation of all future consumption costs, the maximum of all future demand

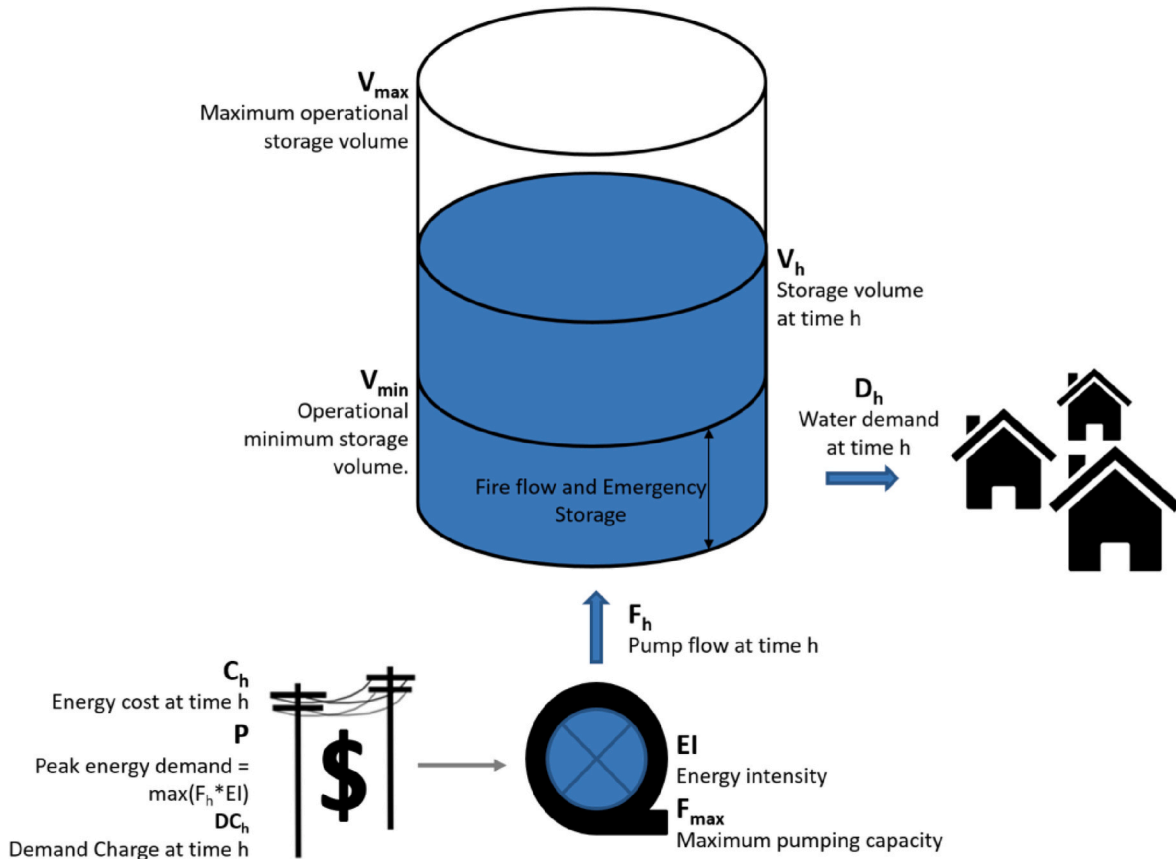


Fig. 1. Components of simplified single utility model with a single water demand, energy demand, and pumping and storage capacity.

charges, and the accumulation of any penalties from constraint violations as shown in Equation (2). An indicator function was used to apply a steep penalty charge if the potential pumping scenario exceeded one of the constraints such as max or min storage volume and max or min pump flows. The penalty cost variables in Equation (2) were used to exclude any potential scenarios that did not adhere to the physical limits of the water system and to ensure that the tank volumes at the end of the optimization run were at or above the starting levels. For each time-step, for each possible pump flow, the best future decision is found by selecting the pumping rates and timings that resulted in the minimum future total cost. This process is repeated recursively until the current time-step, at which point the optimum pump rate and timings are known deterministically. This approach will produce reliable, optimum decisions for an explicit analysis period when all inputs are known, including the future water sales and energy price patterns.

Additionally, we ran a similar optimization to minimize total GHG emissions instead of costs. This slightly changed the objective function so that hourly emissions intensity (E_h) is substituted for consumption cost (C_h) in Equation (2) while also applying a zero value demand charge (DC_h). This process is meant to show how water utilities may change their operations given different objective functions outside of existing pricing signals. The same optimization process and constraints were applied to the GHG scenarios as were applied in the pricing scenarios.

$$\min \left[\left(\sum_{h=1}^H (F_h * EI * C_h) + (X * \mathbb{I}_A) \right) + \max_{h=1}^H \left| (F_h * EI * DC_h) + (X * \mathbb{I}_B) \right| \right]$$

$$\mathbb{I}_A(F_h) = \begin{cases} 0 & \text{if } 0 \leq F_h \leq F_{max} \\ 1 & \text{if } F_h < 0 \text{ or } F_h > F_{max} \end{cases}$$

$$\mathbb{I}_A(V_h) = \begin{cases} 0 & \text{if } V_{min} \leq V_h \leq V_{max} \\ 1 & \text{if } V_h < V_{min} \text{ or } V_h > V_{max} \end{cases}$$

$$\mathbb{I}_B(V_h) = \begin{cases} 0 & \text{if } V_{h=H} \geq V_{h=1} \\ 1 & \text{if } V_{h=H} < V_{h=1} \end{cases} \quad \text{Equation 2}$$

Where:

EI = Pump energy intensity (MWh/MG).

C_h = Energy cost (\$/MWh) during hour h .

E_h = Emission Intensity (mTCO_{2e}/MWh) during hour h .

DC_h = Peak energy demand charge (\$/MW).

X = Penalty charge (~\$Infinite-1).

Optimizations were generated for a two-week long simulation period in each of the 12 months of the year 2020 in hourly intervals. The optimization results were then scaled up to the full 2020 year for each scenario and for each water utility. The two-week optimization time-frame was selected to be tractable from a modeling and computational perspective while also allowing for the impact of weekend and weekday rates and pumping and storage planning for future demands. Optimization time periods longer than two weeks did not yield significantly different results.

3.4. Scenarios

To understand the energy demand flexibility of the water sector and evaluate optimum energy cost operations, multiple signals were assessed as unique scenarios. A baseline, that has a flat consumption rate and no demand charge will establish the flexibility of specific water systems and the overall water sector when introduced to more complex motivational price signals (scenarios 1–8). The potential for energy flexibility was calculated as the difference between the simulated energy demand in the baseline scenario and each of scenarios 1–8. Table 1 summarizes the various rates under each scenario for reference. Additional details about each scenario are available in the SI section.

For real-time pricing (wholesale electricity prices) and emissions

Table 1

Summary of the consumption and demand rates for eight electricity rate and emissions structures.

Scenario	Consumption Rate (\$/kWh)	Demand Rate (\$/kW)	Description
Baseline	Flat	None	Represents 'baseline', i.e. the time and the rate at which a utility consumes energy does not affect the bill. Minimization of costs is only applied to the minimization of total energy demand and not to the time of energy use
1	Flat	Flat	Neither consumption nor demand charge vary with time of day.
2	Flat (Low)	TOU (High)	Very low constant consumption rates with a relatively high demand charge that increases during the afternoon in the summer and early evening in the winter.
3	TOU (Legacy)	Flat	Legacy TOU peak consumption hours are from 12pm to 6pm in the summer. These TOU structures are widely used, but being phased out and replaced with more modern TOU schedules (such as scenario 5).
4	Flat	TOU	Similar in structure to scenario 2, this scenario has relatively high flat consumption charges and lower time dependent demand charges.
5	TOU (Modern)	None	Modern TOU peak consumption hours are from 4 to 9pm when the statewide net electricity grid experiences a ramping of demand.
6	TOU (RTP)	None	RTP represents the CA independent system operator (ISO) locational marginal pricing (LMP) for Northern or Southern CA utilities.
7	Grid operating marginal emissions	None	Optimized based on the marginal emissions intensity for Northern or Southern CA water utilities.
8	Grid average emissions	None	Optimized based on the average emissions intensity of the California Electric grid, estimated for 2020–2030 (CEC, 2020).

scenarios 7 and 8, which will vary for every hour of the year and by location, we chose to represent the diversity of the signals without substantially increasing the complexity of the input data by grouping the water systems into Northern and Southern California. These regions generally correspond to the service territories of Pacific Gas and Electric (PG&E) in the North and Southern California Edison (SCE) and San Diego Gas and Electric (SDG&E) in the South. Any water systems outside of these service territories were assigned to one of these regions based on their geographic proximity to the region. For RTP we utilized day-ahead market locational marginal pricing (LMP) data from the California Independent System Operator in 2020. Prices for the northern region were represented by PG&E Demand Load aggregation point (DLAP) and prices for the southern region were represented by the average of SCE and SDG&E DLAPs. For our emissions optimization scenario, we utilized marginal operating emission rate data from California's Self Generation Incentive Program (SGIP). Similarly to the LMP data, we used the PG&E SGIP data for our northern region, and an average of the SCE and SDG&E SGIP data for our southern region.

The scenarios considered in this analysis represent pricing schedules

currently provided by Southern California Edison and San Diego Gas and Electric with a variety of consumption and demand combinations and structures that are applicable to water utilities with significant pumping demands as well as two scenarios representing average and marginal electric grid GHG emission intensities. Fig. 2 graphically displays the consumption and demand charges for each scenario for the summer and winter by hour of day. For these scenarios summer runs from June 1st until September 30th. Subplots 'A' and 'B' show the various consumption rates under each pricing scenario for each hour of the day on a weekday with subplots 'C' and 'D' showing the weekend rates. Subplots 'E' and 'F' show the demand rates. For scenarios 7 and 8, the consumption rate varied for every hour of the year in our optimization, but follow a similar pattern by season. In Fig. 2, scenarios 6 and 7 are plotted as averages for the winter and summer to demonstrate the typical pattern they followed, although any given hour used in the optimization may have varied from this pattern. Scenarios 7 and 8 emissions intensities were minimized as if they were consumption charges using mTCOe/MWh marginal emissions intensity values as if it was a \$/kWh consumption charge. At any given hour, marginal emissions varied between the northern and southern region, but on average they were the same, which is why you see only one line for marginal emissions in Fig. 2.

3.5. Model outputs and metrics

Several metrics will be used to assess the system performance under the various scenarios as compared to the baseline including total energy consumption, emission intensity, net load shifted, and reduction of the net demand peak. These metrics assess the pricing signal's ability to impact the electric grid when applied to Californian water distribution systems.

3.5.1. Total energy consumption

Assuming constant pumping efficiencies, it takes a certain amount of energy to deliver a given volume of water and we would expect each scenario to consume roughly the equivalent total quantity of energy over the same period. The optimization can result in water systems storing excess water at the end of some simulations in preparation for future time-steps. As a result, some combinations of water system characteristics and energy price signals can result in select water systems potentially ending the simulation with slightly more energy consumed by storing excess water in their storage tanks. This is accounted for by normalizing later results by this total energy consumption.

3.5.2. Net load shifted

Between the hours of 8am – 6pm there can be times where forecast renewable generation exceeds energy demand resulting in curtailment

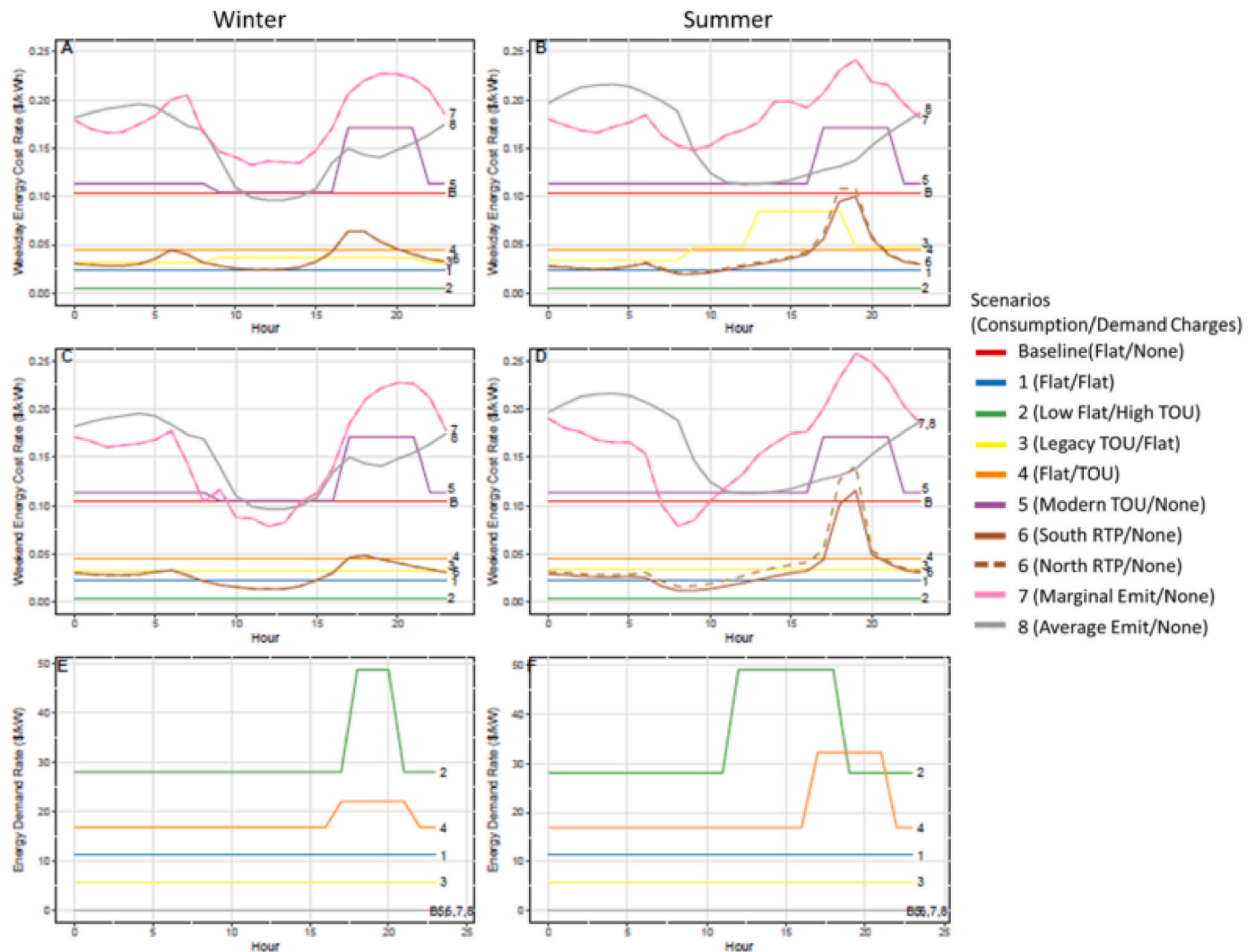


Fig. 2. Energy pricing rate structures (consumption and demand) under each scenario by season, weekdays, and weekends. (Marginal Emissions, scenario 8, are plotted as mTCOe/MWh/2 for ease of plotting).

of renewable resources as shown in Fig. 3 (CAISO, 2021b). The greatest curtailment in CA occurs in the first half of the year, peaking in April with high solar availability and low energy demands and lowest in August with high energy demands resulting in less need for curtailment. In order to avoid curtailment, increase integration of renewable resources, and take advantage of low carbon, often inexpensive, energy sources it may be desirable to shift energy demand to when curtailment is at its highest. Currently in CA the greatest curtailment of renewable resources and the lowest net energy demands occur between 9am and 4pm after which net demand begins ramping up toward an evening peak between 7 and 8pm. We quantify the magnitude of energy shifted into the curtailment window minus any energy shifted out of the 9am – 4pm window as the Net Load Shifted.

3.5.3. Average emission intensity, and avoided emissions

Shifting the energy load may impact the average emission intensity of the water distribution sector given the variable GHG emissions of the electric grid at different times of the day due to the availability of renewable resources (Miller et al., 2022). Total GHG emissions and average emissions intensity will be calculated as in Equation (3) by multiplying the energy profiles under each scenario by emissions factors according to the appropriate hour and month and then summing the emissions and dividing by the total energy consumption. Average emissions intensities highlight the fraction of the grid-wide emissions that can be attributed to the energy consumption (which comes from a variety of sources) of the water distribution sector. The average GHG emissions factors by hour and month for California used in this analysis can be found in the SI section.

We also considered marginal emissions intensities because they better represent the consequential impact of load shifting and avoided emissions. We calculated avoided emissions under each scenario by multiplying the change in hourly electricity demand (baseline minus scenarios 1–8) by hourly marginal emissions factors as in Equation (4). This accounts for the emissions intensity of the last unit of energy that

would have been produced to meet demand. By reducing demand when marginal emissions intensities are high, the last units of energy are not required and those emissions are avoided. Avoided emissions account for the impact of shifting water distribution demands on the electric grid (which changes the last unit of produced energy), i.e. how the water sector can change overall grid emissions.

$$\text{Average Emission Intensity} \left(\frac{m\text{TCO}_2e}{MWh} \right) = \frac{\sum_{t=0}^T (\text{Energy}_t * \text{AvgEmissionFactor}_t)}{\text{Total Energy}} \quad \text{Equation 3}$$

$$\text{Avoided Emissions} (m\text{TCO}_2e) = \sum_{t=0}^T \left[(\text{Energy}_t (\text{Baseline}) - \text{Energy}_t (\text{Scenario } x)) * \text{MargEmissionsFactor}_t \right] \quad \text{Equation 4}$$

3.5.4. Net demand peak reduction

Net demand peak is the maximum net energy demand (total energy demand minus wind and solar generation) in a single interval each day. This quantity represents the total nonrenewable capacity that was required that day and can dictate the number and size of fossil fuel generated power plants required to meet energy demand in California (CEC, 2021). Reducing the net energy demand peak may mean greater grid reliability as the grid can have greater capacity to handle demands and reduce the likelihood of rolling blackouts or insufficient energy supplies during high demand periods. A reduction in the net demand peak may also mean reducing grid emissions since peak demand is often served by less-efficient and more emitting ‘peaking power plants’ (Torriti, 2015). Regular peak energy demand has been shifting later in the day since 2011 with the summer total demand peak occurring on average around 6pm in 2020 and the net demand peak occurring after 7pm (CEC, 2021). This metric of impact will be assessed at the statewide

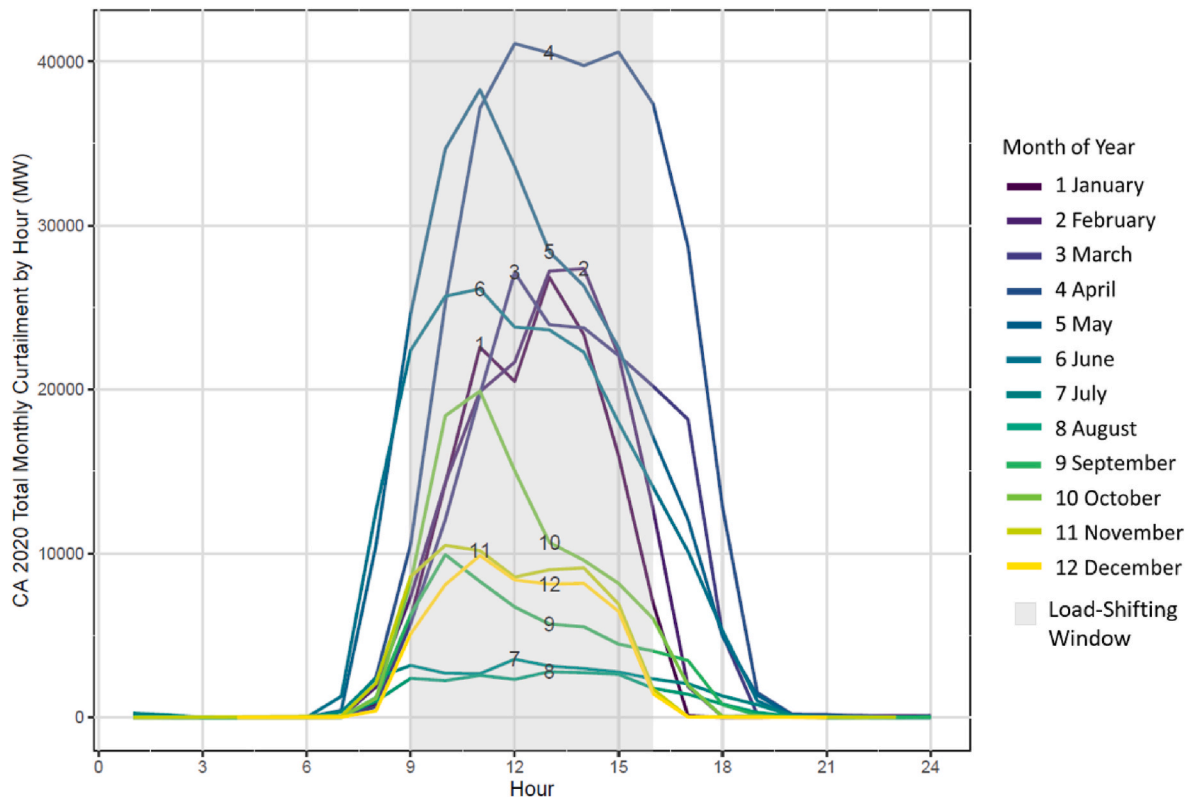


Fig. 3. Total Solar and wind curtailment by hour of day and month in California for 2020 (Data from CAISO, 2021a).

level by examining the overall water distribution sector net energy demand curve of the baseline relative to the different scenarios at 7pm. For perspective, in 2020 the annual California grid-wide instantaneous peak demand occurred in August at 47,121 MW with typical daily net demand peaks throughout the year around 30,000 MW (CAISO, 2021b).

4. Results

4.1. Utility level

At the individual utility level, we saw that the ability to load shift was largely limited by storage volume relative to water demand, the total

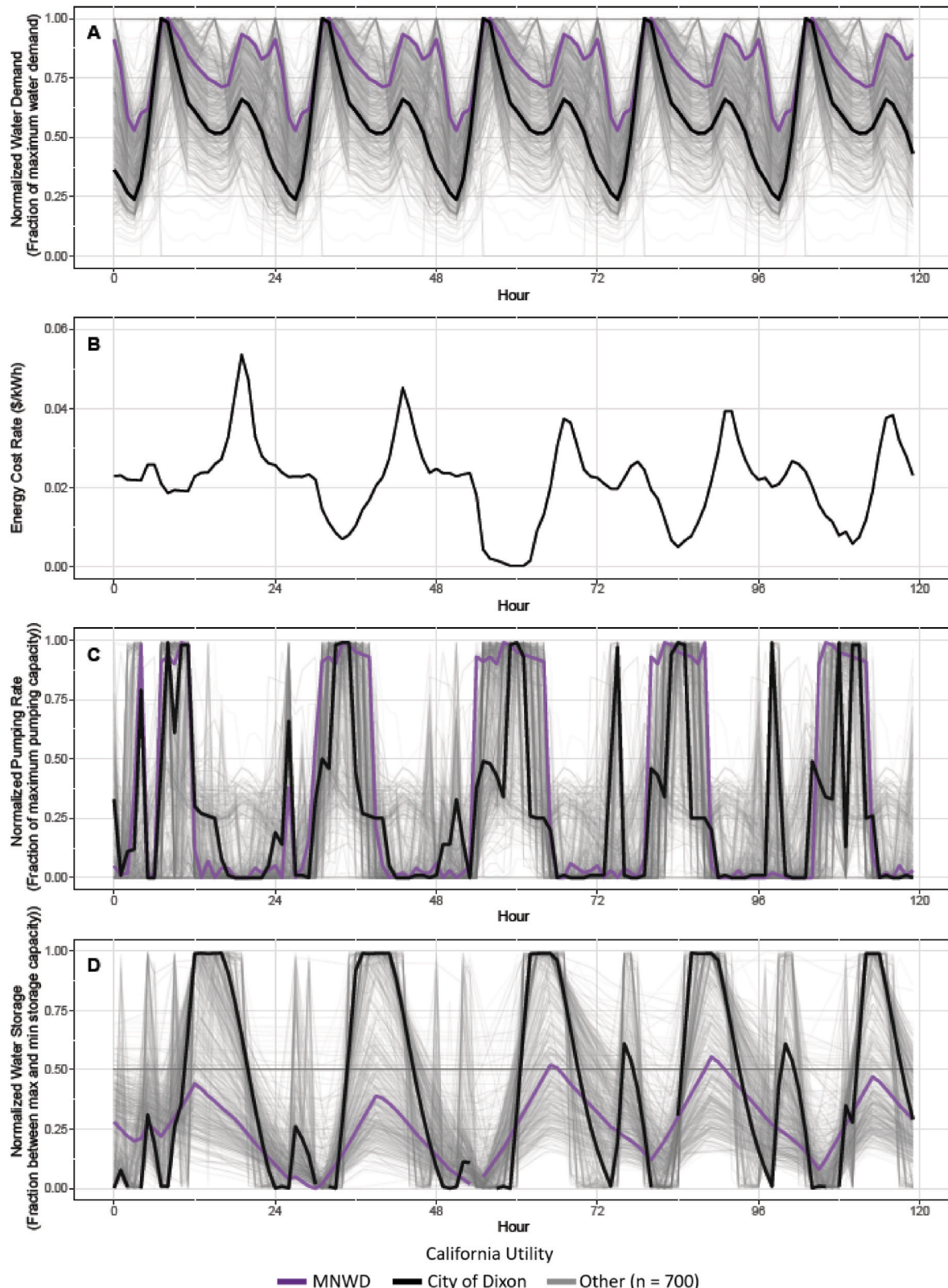


Fig. 4. Optimized operations for individual utilities for five days in May starting on a Friday under real-time pricing scenario 6.

pumping capacity, and local minima in the pricing structures. In order to illustrate the outputs of our modeling, before reporting the aggregated statewide results, we first focus on two water utilities with different characteristics under the real-time pricing (RTP) scenario 6. The first water utility, plotted in purple on Fig. 4, is the Moulton Niguel Water District (MNWD) in Southern California, which is characteristic of larger water systems with large amounts of storage relative to their water demand. Plotted in black is the City of Dixon in Northern California, which is characteristic of a smaller water system with less available storage relative to water demand.

The ability for an individual utility to load shift is limited by their storage volume. In Fig. 4, during the relatively high costs of energy near 7pm (panel 'B'), the MNWD optimization avoided pumping (panel 'C') and used sizeable storage (panel 'D') to meet water demand until energy costs decreased. On Sunday (hours 48–72) energy costs in the afternoon were at their lowest for the week, reaching as low as \$0/kWh. MNWD took advantage of Sunday's low rates and used their storage to pump almost constantly throughout the afternoon until prices increased in the evening when they were able to shut off their pumps and meet water demand by draining their stored water. During this same period, the City of Dixon also increased their pumping, but could not take advantage of the entire low pricing period given their limited storage. Panel 'D' of Fig. 4 shows Dixon filling their tanks to capacity when energy prices are low and draining to their minimum when prices increased. We noted numerous utilities, shown as thin gray lines in Fig. 4, maxing out their tanks during low pricing periods indicating that their storage was limiting their ability to load shift.

Total pumping capacity also plays a role in the ability of an individual utility to load shift. MNWD has plenty of storage capacity, but at the global minimum price signal their pumps could not completely fill their tanks and instead they had to pump later in the week when prices were at a local low. We see this shown in Fig. 4 around hour 60, when prices reached \$0/kWh and most utilities pumped at full capacity (provided they had storage), but most could not sufficiently fill their tanks during this time period and had to pump again around hour 75 (early Monday morning) during a local low price, but outside the load-shifting window.

Most utilities will shut off their pumps when facing relatively high price signals, but not all utilities have sufficient storage or pumping capacity to do this. In Fig. 4 we see the RTP signal peaking around hour 19 (7pm) and for several utilities (thin gray lines) their pumping rate in panel 'C' remains high or even increases during this time in order to deliver adequate water volumes to meet demand. In cases where a water system has no appreciable storage, they are forced to pump to exactly meet demand and therefore demonstrate no capacity for load-shifting.

Global minima and maxima prices matter, but local minima and maxima in rate structures can be especially impactful on utilities that have limited storage or pumping capacity. As seen in panels B and C of Fig. 4, both MNWD and Dixon pumped when local lows in the pricing scheme occurred. In the case of Dixon, they did not have adequate storage to meet demand for extended periods and were forced to pump whenever a local low energy price occurred even when a lower price would be available in a few hours. In the case of MNWD, which has excess storage, they could be more selective on when they pumped and could wait for lower pricing periods, but due to their pumping limitations they could not fully take advantage of their storage capacity and the global minima and were still impacted by local minima prices.

4.2. Statewide level

Summing the energy demands for the 702 water distribution systems in our dataset resulted in an annual energy usage of approximately 3200 GWh for distribution pumping, shown in Table 5, making up roughly 1.2% of the 272,576 GWh total electrical energy consumed in California in 2020 (CEC, 2020). Since the water systems modeled represent 84% of the water distribution in California, we assume that the entire urban

water distribution sector in California uses approximately 1.4% of the total statewide electricity consumption.

The aggregated optimization results across the eight scenarios yields insight into the ability of the water sector as a whole to impact the California electric grid and statewide energy and emissions goals. Depending on the scenario, utilities behaved differently to minimize cost or minimize GHG emissions resulting in varying energy demand profiles, electric grid impacts, and load shifting potentials.

4.2.1. Net load shifted

Non-demand charge scenarios (5–8) resulted in significant shifts in energy loads into the curtailment window in aggregate over the entire year. Fig. 5 plots the annual daily average energy demand in California by the water distribution sector by hour of day under each optimization scenario. Marked on the figure as a shaded gray area is the curtailment window representing the hours when sizable quantities of energy are curtailed in California along with a dotted line representing the actual curtailment values for 2020. Scenarios with demand charges (1–4, solid lines) all resulted in minimal or negative load shifting and relatively flat hourly consumption rates so as to avoid being penalized for periods of high consumption. Scenarios 5–8, without demand charges (dashed lines), all resulted in positive load shifting and much more dynamic energy consumption over a 24-h period. These patterns held true for both the winter and summer periods as shown in panels 'A' and 'B' of Fig. 6.

Pricing signals featuring their lowest pricing or emissions rates during the curtailment window with relatively high rates at all other times (scenarios 5&8) resulted in the greatest load-shifting. Scenario 8 which minimized average GHG emissions resulted in the greatest annual net load shifted with the ability to shift approximately a third of its energy demand, approximately 1070 GWh, followed by scenario 5 with a modern TOU energy consumption rate without a demand charge. Scenarios 6 (RTP) and 7 (Marginal Emissions) also resulted in significant load-shifting, but to a lesser degree due to local low rates in the early morning outside the load-shifting window.

4.2.1.1. Impact on curtailment. If we consider net load shifted as the potential to reduce curtailment of renewable energy, i.e. increase the capacity for integration of renewable energy, then statewide curtailment could be reduced by up to 68% under the optimized average emission scenario (scenario 8). In Fig. 5, California's average daily renewable energy curtailment for 2020 is plotted by hour of day relative to the energy demand by the water sector under the various scenarios. In 2020 the California Independent Service Operator curtailed a total of 1580 GWh of solar and wind energy (CAISO 2021a).

4.2.1.2. Seasonal impacts. The capacity for utilities to load-shift is greater in the winter than in the summer. Of the scenarios that were successful at load-shifting, with the exception of the average emission scenario 8 which performed similarly in summer and winter, their capacity to do so more than doubles in the winter months as shown in Table 3 and Fig. 6. One reason for this could be the different energy pricing schemes for summer and winter. For scenario 5 (modern TOU) this could be a result of a dip in energy prices during the load-shifting window in the winter which does not exist under the summer pricing scheme. For scenarios 6 and 7 (RTP and Marginal Emissions), their low prices are better aligned with the load-shifting window during the winter months whereas they skew earlier during the summer (see Fig. 2). The second reason for this is reduced water demand in the winter (largely due to reduced irrigation demands) paired with constant storage and pumping capacity allowing for greater flexibility to shift loads.

4.2.2. Total GHG emissions, average emission intensity, and avoided emissions

Shifting pumping to the middle of the day means lower overall GHG

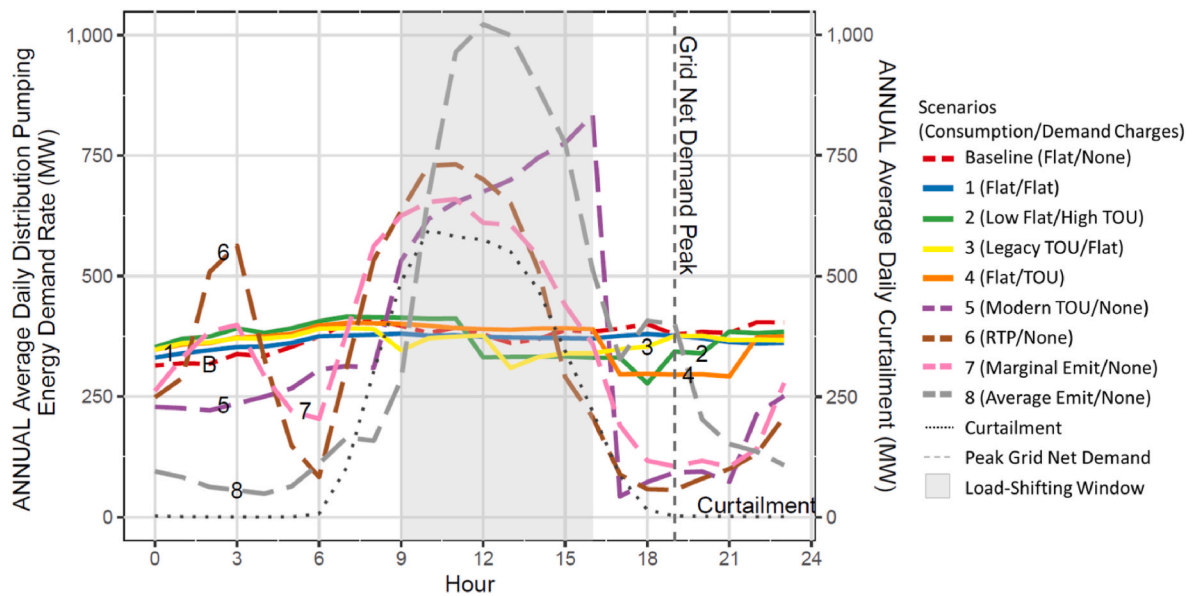


Fig. 5. Aggregated annual daily average water sector pumping energy demand by hour of day under eight optimization scenarios, CA 2020 electric energy curtailment by hour of day, and 2020 CA net demand peak hour. Solid lines indicate the presence of a demand charge.

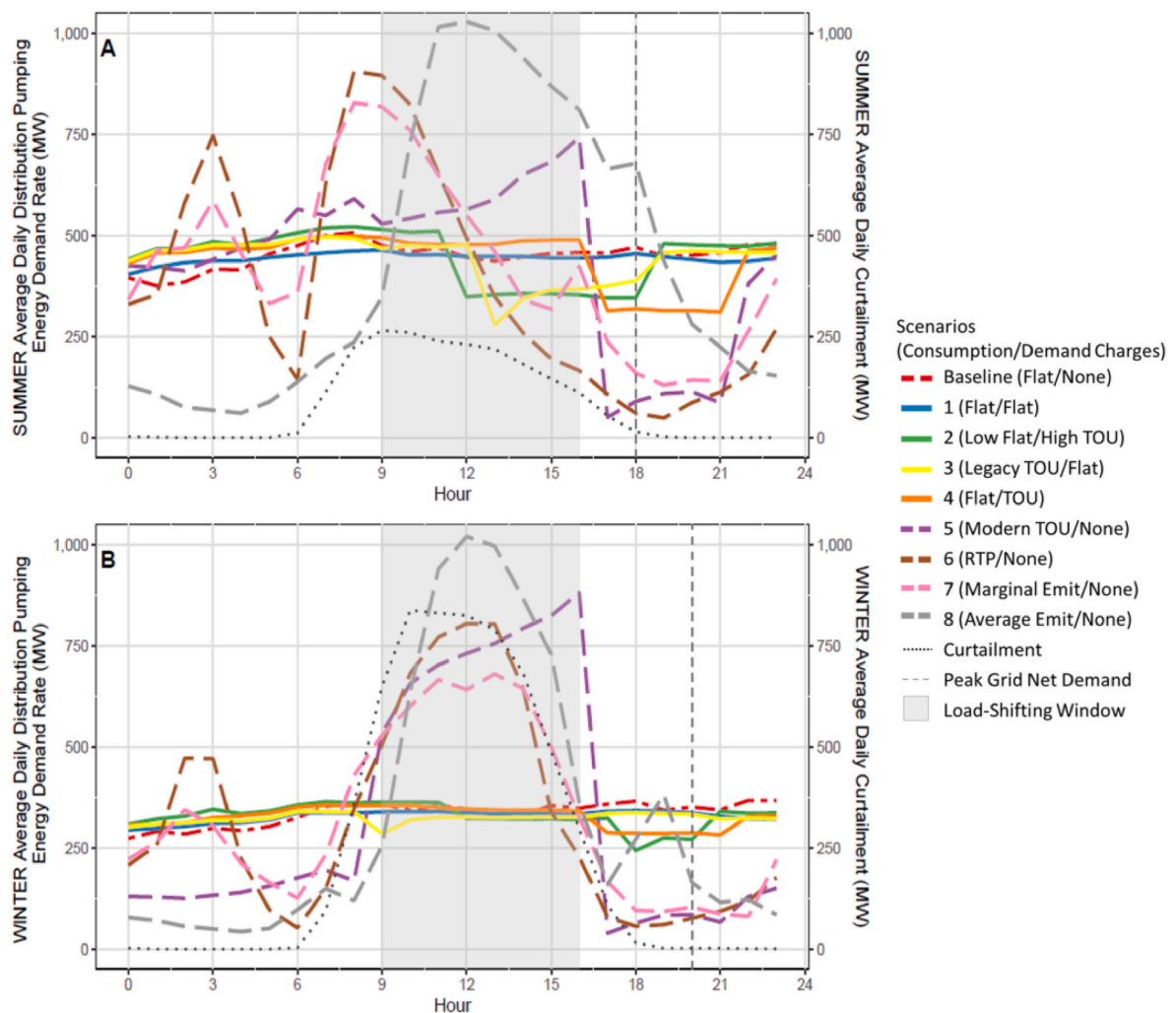


Fig. 6. Aggregated A) summer and B) winter daily average water sector pumping energy demand by hour of day under eight optimization scenarios, CA 2020 electric energy curtailment by hour of day, and 2020 CA net demand peak hour. Solid lines indicate the presence of a demand charge.

emissions due to the prevalence of solar energy on the California electric grid. As expected, and shown in Table 2, operating to minimize average GHG emission under scenario 8 resulted in the lowest average emission intensity and reduced water-sector emissions by 19% as compared to the baseline. Other scenarios that shift load into the middle of the day (scenarios 5–7), when grid carbon intensity is lower, also reduced emissions compared to the baseline. Although the RTP scenario shifted load into low carbon intensity periods, it also increased energy consumption during the early morning, when carbon intensity is high, resulting in a low net reduction in water sector emissions.

Operating to minimize indirect avoided grid emissions using a marginal emissions signal, unsurprisingly, results in the greatest quantity of avoided GHG emissions, followed closely by the RTP scenario. Although the modern TOU and average emission optimization scenarios resulted in the largest load shifting and reduction in renewable energy curtailment, they avoided less than half of the emissions avoided by the RTP and marginal emissions scenarios. However, while these results describe the short-run emissions impact of load shifting, these results do not describe how more permanent load shaping in the water sector might affect long-run avoided emissions.

4.2.3. Net demand peak reduction

RTP, modern TOU consumption, and marginal emissions (scenarios 6, 5, and 7) resulted in the largest reductions in the net demand peak, 85.1%, 75.6%, and 72.0% over baseline, respectively. This equates to reducing the overall grid peak by approximately 1%. Table 4 summarizes annual average daily net demand peaks under each scenario for comparison.

Table 2

Annual total energy consumed, shifted, emissions, and emission intensity under eight different optimization scenarios.

Scenario (Consumption/Demand Charges)	Annual Total Energy Consumed (MWh) ^a	Annual Net Load Shifted into Midday (MWh) ^b	Average GHG Emission Intensity (mTCO2e/MWh) ^c	Annual Total Avoided Emissions (mTCO2e) ^d
Baseline (Flat/None)	3,273,163	0.0	0.1552	0.0
1 (Flat/Flat)	3,214,285	−16,660	0.1560	21,859
2 (Low/High)	3,228,971	−37,354	0.1578	29,200
3 (Legacy TOU/Flat)	3,183,231	−80,584	0.1579	34,005
4 (Flat/TOU)	3,202,021	30,195	0.1559	42,621
5 (Modern TOU/None)	3,189,280	740,900	0.1416	149,178
6 (RTP/None)	2,989,939	578,158	0.1518	289,574
7 (Mar. Emit/None)	3,134,091	535,682	0.1488	330,627
8 (Avg. Emit/None)	3,179,347	1,071,478	0.1263	128,256

^a Total energy consumption under all scenarios should be identical given identical water demands. Small variations exist because each optimization scenario may have resulted in slightly different final water storage levels each month resulting in slightly more or less total energy being demanded for pumping. The total energy consumed for pumping under all scenarios equates to approximately 1.2% of the total energy consumed in CA in 2020.

^b Net load shifted is relative to the baseline considering the energy consumed between the hours of 9am and 4pm. Negative values indicate that a greater load was shifted out the load shift window than in.

^c Average GHG emission intensity, calculated as the annual sum of the energy demand in each hour multiplied by the corresponding hourly average CA electricity grid emissions rate divided by the annual total energy consumed as reported in the first column.

^d Avoided emissions, calculated as the sum of the differences in energy demands between scenarios 1–8 and the baseline scenario multiplied by the operating marginal emissions intensity of the CA electric grid for every hour in 2020.

Table 3

Annual total energy consumed, shifted, emissions, and emission intensity under eight different optimization scenarios.

Scenario (Consumption/Demand Charges)	Summer Daily Average Net Load Shifted (MWh) ^{a,b}	Winter Daily Average Net Load Shifted (MWh) ^a
Baseline	0	0
1 (Flat/Flat)	−32	−53
2 (Low Flat/High TOU)	−239	−34
3 (Legacy TOU/Flat)	−307	−178
4 (Flat/TOU)	196	26
5 (Modern TOU/None)	926	2584
6 (RTP/None)	482	2137
7 (Marg. Emissions/None)	706	1849
8 (Avg. Emissions/None)	2749	3029

^a Net load shifted is relative to the baseline considering the energy consumed between the hours of 9am and 4pm. Negative values indicate that a greater load was shifted out the load shift window than in.

^b Summer represents the months of June through September.

5. Discussion

The results from this experiment support three major findings: 1) the water sector currently has the physical potential to play a role in supporting a resilient and reduced carbon intensity electricity grid, 2) energy rate structures drive the water sectors capacity to influence the electric grid, and 3) in a changing energy market, energy rates must change to reflect modern goals.

5.1. Water Sector's potential as a grid resource

Shifts to water distribution pumping will not solve California's energy concerns by itself, but its potential impact is not negligible when compared to existing policies and incentives. The results of this work show that the water sector could shift up to one third of its total energy demand from undesirable periods into the middle of the day to promote increased integration of solar energy. Given that the water distribution sector accounts for 1.4% of the total electricity consumption in California, this could mean potentially moving almost 0.5% (1275.6 GWh) of California's total electricity demand into the middle of the day using already built infrastructure and no changes to the consumption patterns of individual water users. Additionally, our modeling suggests that the water sector could reduce the net demand peak by up to 321 MW or approximately a 1% reduction of the overall net demand peak of the grid (CAISO, 2021b). For perspective, electricity utility demand response programs combined statewide accounted for 3–4% of the grid adequacy

Table 4

Annual average daily peak net energy demanded from the CA water distribution sector, magnitude and percent reduction in average daily net demand peak relative to scenario one for all eight scenarios.

Scenario (Consumption/Demand Charges) ^a	Annual Average Daily Peak Net Energy Demanded (MW) ^a	Change in Annual Average Daily Peak Net Energy Demand relative to baseline (MW)	Percent Change in Average Daily Peak Net Demand relative to baseline (%)
Baseline	378.3	0.0	0.0
1 (Flat/Flat)	376.1	−2.1	−0.6
2 (Low/High)	342.5	−37.9	−9.3
3 (TOU/Flat)	375.2	−2.6	−0.7
4 (Flat/TOU)	295.3	−82.7	−21.8
5 (TOU/None)	92.5	−285.7	−75.6
6 (RTP/None)	56.3	−321.5	−85.1
7 (Marginal Emissions)	105.6	−272.1	−72.0
8 (Average Emissions)	399.2	21.4	5.7

^a Peak energy measured as energy demanded at the 7 pm hour (representing the approximate net demand peak on the CA grid in 2020).

Table 5

Best and worst electricity rate structures under a given metric as it pertains to shifting energy demands in the California water distribution sector.

Metric	Best Rate Structure	Worst Rate Structure
Shifting Energy	8 (Average Emissions)	3 (Legacy TOU Rate)
Avoiding Emissions	7 (Marginal Emissions)	2 (High Demand Charge)
Reducing the Net Demand Peak	6 (Real-Time Pricing)	8 (Average Emissions)
Overall (combined metrics)	5 (Modern TOU)	4 (Legacy TOU)

capacity in the summer of 2021 (CAISO, 2022).

Shifting water distribution pumping not only allows for greater integration of more renewable resources and greater load capacity, but it could also avoid GHG emissions by as much as 330,627mTCO₂e/yr, or approximately 0.55% of the total emissions of the California electric grid in 2020 (California Air Resources Board, 2022). Under California cap-and-trade markets this reduction could be worth as much as \$US 5.6 M (based on Q4 2020 Prices of \$16.93/mTCO₂e, CARB, 2023).

5.2. The role of energy rate structures in the water sector

For the water sector to meet its potential as a grid resource, water utilities need to be appropriately incentivized. This means removing electricity demand charges, matching the TOU consumption charges and timings to grid goals, and being aware of water system limitations with regard to water storage capacity, pumping, and water demands.

All scenarios with demand charges resulted in relatively flat energy consumption and low potential to load-shift or reduce peak demands. Demand charges discourage utilities from high levels of energy consumption at any given time thus resulting in utilities maintaining a consistent level of consumption to minimize their overall demand. Removing demand charges allows utilities to respond to dynamic pricing signals and be more responsive overall to the needs of the grid.

Not every water utility can participate in load-shifting, but when faced with high energy prices (in the absence of a demand charge) the water sector as a whole was able to reduce pumping until prices dipped back down. If those pricing signals correspond to the load shifting window or a net demand peak (i.e. lowest prices between 9am and 4pm and highest between 5 and 9pm) utilities are able to perform significant amounts of load-shifting into the middle of the day and away from the early evening. To maximize this, the pricing signals need to be greater at all times of the day outside the load shifting window. Several of the rate structures we examined do this to an extent, but inconsistently with occasional low prices during periods of high carbon intensity and/or outside the load-shifting window. The modern TOU rate structure we examined did this during the winter, which is what allowed it to be so successful at load shifting during this period.

Local minima in a TOU rate structure can have a significant impact on energy consumption in the water sector. Most water utilities cannot wait to pump until the daily minimum prices occur due to storage and pumping constraints. Even when incentivized to pump in the middle of the day, they were forced to also pump at alternative times, often in the early morning when local minimum prices occurred. The most obvious example of this is with scenarios 6 and 7 which have local minima around 3am. This resulted in less energy being shifted into the middle of the day, but greater avoided emissions. Timing of the local minima play a significant role in how water utilities respond to the pricing signals and their ability to meet electricity grid goals.

In general, the modern TOU pricing structure (scenario 5) performed consistently at the top for all of the metrics we examined. Other scenarios, like the average emissions rate did well at shifting energy and avoiding emissions but performed poorly when considering grid capacity as related to the net demand peak. Depending on the goal, different energy rate structures performed the best. Table 5 summarizes

the best and worst performing rate structures by the metrics used in this paper.

5.3. Modern energy markets need modern rate structures

Scenario 3, which has a legacy TOU consumption charge, resulted in the largest amount of load shifted out of the curtailment window. Legacy TOU consumption rates are still widely used by many electricity utilities and have their lowest prices during the night when energy demand is typically at its lowest. These rates were primarily designed to shift energy demand away from peak demand periods to limit the overall energy demand on the grid at any point in time.

These rates made a lot of sense when the energy generated during the night and the day both came from the same fossil fuel-based sources and resulted in the same emission intensities, which is no longer true. These rates which used to make energy grids more resilient are breaking down in a new energy market. When applied to the water sector, legacy TOU rates resulted in grid impacts working counter to their original goals, demonstrating the need to remove and replace them with rates reflective of our changing energy market.

5.4. Limitations

While we believe that this study represents the best available estimate to date of the demand flexibility potential of the California water sector, it has several important limitations. Water distribution systems are complex and this method simplifies utility operations in order to account for the 700+ public water systems in California. In the simplification of water systems, important operational constraints have been ignored, including water quality requirements, pressure requirements, and forecasting ability.

Further research with more detail could improve the results and provide additional insight about the application of the principles discussed in this paper. Additionally, RTP and emissions rates are dynamic and would change in response to water system load shifting and purchasing energy at alternative hours. If the entire water sector were to participate in the wholesale market (i.e. RTP) that may change the energy rates and this analysis. Likewise, more renewables being added to the grid as is expected to meet CA renewable energy goals, will influence the energy rates as well as the emissions intensities.

For the purposes of our study we assumed a baseline energy rate tariff with flat energy consumption rates and no demand charges. In reality, water utilities are likely to be using a variety of energy rate tariffs, often multiple at a single utility. The purpose of assigning a baseline was to indicate the flexibility of the water sector, i.e. how much could the water distribution sector shift its energy demands when presented with different energy rate tariffs. A flat rate tariff was selected as the baseline because it results in a flat demand profile and operates independent of time (i.e. no matter how or when a utility consumed energy the total bill would be the same). Any of the scenarios could be used as a baseline and numbers recalculated using the values presented in this paper.

Our optimization assumed perfect forecasting ability, which would not be present in the real world. Being able to perfectly forecast water demands and dynamic energy prices give a water system operator the ability to plan days ahead for low or high upcoming prices or water demands allowing for a minimization of cost while ensuring reliability. In reality, under some of the scenarios, water systems might not be aware of low energy prices until a few hours ahead, which will change how and their ability to shift their pumping demands. This limitation especially applies to dynamic energy tariffs such as wholesale (RTP) pricing and emissions minimization scenarios. For this reason, our estimate may be on the high side relative to a real-life operator who may be more cautious in the face of uncertainty.

5.5. Future exploration and what is needed

From the findings of this research, there are several opportunities for extensions or additional exploration to be undertaken. A pilot application of a handful of utilities with the ability to respond to dynamic energy prices such as RTP could be useful in validating the results and the flexibility of the water sector to perform load-shifting. Further, using what was uncovered in this research to create an 'optimum' rate structure for water utilities and applying the same or similar optimization process could help clarify the full potential, not limited to the currently available rate tariffs, of the water distribution network in California to respond to grid needs. Lastly, this research largely focused on the impacts of load-shifting in the water sector on the electric grid, but did not explore the impacts to the distribution systems themselves such as costs, water quality, operational challenges, etc. Future research may consider what is occurring on the individual utility level and how that affects water distribution goals and operations. This research could consider questions such as which rates might be easiest for a utility to adopt, what operational controls are needed to optimize energy demands, and how a rate structure could be specifically designed for the water sector considering any sector specific limitations and challenges.

This analysis focused on reducing curtailment and assigned a load-shifting window to 9am-4pm, but could have been shifted to examine another time period or address other needs of the electric grid. This study did not examine the impact of the various energy rate scenarios on grid concerns such as ramping of energy demands, which can occur leading up to the net demand peak. Additional analysis that examines how the water sector can be leveraged to meet other grid-related goals could be informative.

6. Conclusion

Potable distribution pumping in the water sector accounts for approximately 1.4% of the total electricity consumption in California annually and has the capacity to shift when it uses that energy to help integrate more renewable resources onto the grid, reduce curtailment, decrease GHG emissions, and reduce the net demand peak. We account for the operations of 702 public water distribution systems in California with simulated water demand, pumping, and energy demands under eight different energy pricing and emissions minimization scenarios. From this simulation, we find that the water sector has the flexibility to shift its energy demands by up to 1071 GWh annually, to reduce curtailment in California by up to 68%, to reduce its contribution to peak net demand by up to 321 MW, and/or avoid 330,627mTCO₂e of emissions depending on the energy rate tariff.

The potential distribution system response depends on individual system's combination of tank storage capacity, pumping capacity, and water demands as well as the applied energy tariff structure. Most systems had the ability to perform some load shifting with the greatest potential at systems with high storage volumes, low water demands, and high pumping capacity. Energy tariffs with demand charges resulted in minimal and in some cases negative load-shifting compared to the baseline. The tariffs without demand charges and TOU consumption charges that aligned low pricing with the load shifting window and higher prices outside the window, resulted in the most dynamic energy consumption, the greatest shifts in energy load, and the lowest GHG emissions. Depending on the objective, adjusting rate structures could encourage the water sector to shift when it demands energy such that it could better assist California in meeting future energy and emission targets.

Funding

This study was funded by the California Energy Commission with financing from the Electric Program Investment Charge (EPIC), the ratepayer surcharge authorized by the California Public Utilities

Commission, through agreement number EPC-16-062. Any opinions, findings, and conclusions expressed in this material are those of the authors and do not necessarily reflect the views of the funding agency.

CRedit authorship contribution statement

Amanda M. Rupiper: Conceptualization, Data curation, Formal analysis, Methodology, Visualization, Writing – original draft. **Robert Good:** Conceptualization, Data curation, Formal analysis, Methodology, Software, Writing – review & editing. **Gregory J. Miller:** Conceptualization, Data curation, Formal analysis, Methodology. **Frank Loge:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

A special thanks to student interns Kideok Kwon, Jonathan Fernandez, and Rohan Malhotra who helped with dataset cleaning and processing.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jclepro.2023.140373>.

References

- Aghajanzadeh, A., McKane, A., Wray, C., Therkelsen, P., 2015. 2006-2015 Research Summary of Demand Response Potential in California Industry, Agriculture, and Water Sectors. Lawrence Berkeley National Laboratory LBNL, 1004131.
- Amaral Lopes, R., Gronborg Junker, R., Martins, J., Muta-Pina, J., Reynders, G., Madsen, H., 2020. Characterisation and use of energy flexibility in water pumping and storage systems. *Appl. Energy* 277, 115587.
- Beck, F., Martinot, E., 2004. Renewable energy policies and barriers. *Encyclopedia of Energy* 5.
- California Air Resources Board (CARB), 2022. California Greenhouse Gas Emissions for 2000 to 2020: Trends in Emissions and Other Indicators. https://ww2.arb.ca.gov/sites/default/files/classic/cc/inventory/2000-2020_ghg_inventory_trends.pdf.
- California Air Resources Board (CARB), 2023. Cap-and-Trade Program Data Dashboard: Carbon Allowance Prices. Retrieved on 12/15/2023 from. <https://ww2.arb.ca.gov/our-work/programs/cap-and-trade-program/program-data/cap-and-trade-program-data-dashboard#Figure7>.
- California Energy Commission (CEC), 2021. A peek at net peak. <https://www.energy.ca.gov/data-reports/energy-insights/peek-net-peak>. (Accessed 11 September 2021).
- California Energy Commission (CEC), 2020. 2020 total systems electric generation. <https://www.energy.ca.gov/data-reports/energy-almanac/california-electricity-data/2020-total-system-electric-generation>. (Accessed 11 September 2021).
- California Independent Systems Operator (CAISO), 2017. Impacts of Renewable Energy on Grid Operations chrome-extension:<http://efaidnbnmnnibpcajpcgiclfndmka/j/viewer.html?pdfurl=https%3A%2F%2Fwww.caiso.com%2Fdocuments%2Fcurtailmentfastfacts.pdf&clen=743715&chunk=true>.
- California Independent Systems Operator (CAISO), 2021a. Today's outlook : current and forecasted demand. <http://www.caiso.com/TodaysOutlook/Pages/default.aspx#section-net-demand-trend>. (Accessed 11 September 2021).
- California Independent Systems Operator (CAISO), 2021a. Managing Oversupply. <http://www.caiso.com/informed/Pages/ManagingOversupply.aspx>.
- California Independent Systems Operator (CAISO), 2022. Demand Response Issues and Performance 2021. Department of Market Monitoring. <https://www.caiso.com/Documents/Demand-Response-Issues-Performance-Report-Jan-12-2022.pdf>.
- California State Water Resources Control Board (CA SWRCB), 2023. Electronic Annual Report. Accessed from: https://www.waterboards.ca.gov/drinking_water/certlic/drinkingwater/ear.html.

- Denholm, P., Hand, M., 2011. Grid flexibility and storage required to achieve very high penetration of variable renewable electricity. *Energy Pol.* 39 (3), 1817–1830. <https://doi.org/10.1016/j.enpol.2011.01.019>.
- Faruqui, A., Sergici, S., 2013. Arcturus: international evidence on dynamic pricing. *Electr. J.* 26 (7), 1040–6190. <https://doi.org/10.1016/j.tej.2013.07.007>.
- Ge, M., Friedrich, J., Vigna, L., 2020. 4 Charts Explain Greenhouse Gas Emissions by Countries and Sectors. World Resources Institute. Accessed 12/30/2023. <https://www.wri.org/insights/4-charts-explain-greenhouse-gas-emissions-countries-and-sectors>.
- Kernan, R., Liu, X., McLoone, S., Fox, B., 2017. Demand side management of an urban water supply using wholesale electricity price. *Appl. Energy* 189, 395–402.
- Menke, R., Abraham, E., Parpas, P., Stoianov, I., 2016. Demonstrating demand response from water distribution system through pump scheduling. *Appl. Energy* 170, 377–387.
- Miller, G., Novan, K., Jenn, A., 2022. Hourly accounting of carbon emissions from electricity consumption. *Environ. Res. Lett.* 17, 044073 <https://doi.org/10.1088/1748-9326/ac6147>.
- Oikonomou, K., Parvania, M., Khatami, R., 2018. Optimal demand response scheduling for water distribution systems. *IEEE Trans. Ind. Inf.* 14, 5112–5122.
- Park, L., Minasian, C., Wells, J., Vu, M., 2019. Water Sector Over-generation Mitigation and Flexible Demand Response, Phase 2 Program Recommendations. Water Energy Innovations.
- Robinson, L., Edwards, J., Willnow, L., 2012. Computer modeling of water distribution systems. In: AWWA MANUAL M32, third ed. American Water Works Association, 2012, Denver, CO.
- Sherwood, J., Chitkara, A., Cross-Call, D., Li, B., 2016. A Review of Alternative Rate Designs: Industry Experience with Time-Based and Demand Charge Rates for Mass-Market Customers. Rocky Mountain Institute. http://www.rmi.org/alternative_rate_designs.
- Sowby, R.B., 2021. Increasing water and wastewater utility participation in demand response programs. *Energy Nexus* 1, 2772–4271. <https://doi.org/10.1016/j.nexus.2021.100001>.
- Sowby, R.B., 2023. Making waves: research to support water and wastewater utilities in the transition to a clean-energy future. *Water Res.* 233 (15) <https://doi.org/10.1016/j.watres.2023.119739>.
- Torriti, J., 2015. Peak Energy Demand and Demand Side Response, first ed. Routledge, New York. <https://doi.org/10.4324/9781315781099>.
- United Nations, 2021. The Sustainable Development Goals Report. <https://unstats.un.org/sdgs/report/2021/>. (Accessed 23 May 2022).
- United States Energy Information Administration (US EIA), 2021. California's Curtailments of Solar Electricity Generation Continue to Increase. <https://www.eia.gov/todayinenergy/detail.php?id=49276>. (Accessed 19 April 2021).
- Van Triel, F., Lipman, T.E., 2020. Modeling the future California electricity grid and renewable energy integration with electric vehicles. *Energies* 13, 5277. <https://doi.org/10.3390/en13205277>.
- Wimpler, C., Hejazi, G., de Oliveira Fernandes, E., Moreira, C., Conner, S., 2017. Impacts of load shifting on renewable energy integration. *Energy Proc.* 107, 248–252.
- Zohrabian, A., Plata, S.L., Min Kim, D., Childress, A.E., Sanders, K.T., 2021. Leveraging the water-energy nexus to derive benefits for the electric grid through demand-side management in the water supply and wastewater sectors. *WIREs Water* 8 (3). <https://doi.org/10.1002/wat2.1510>.